

LiDAR SCOUT ROVER

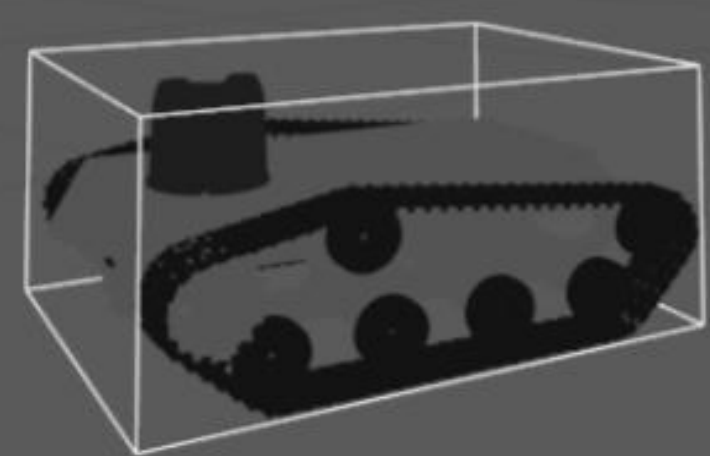
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Introduction

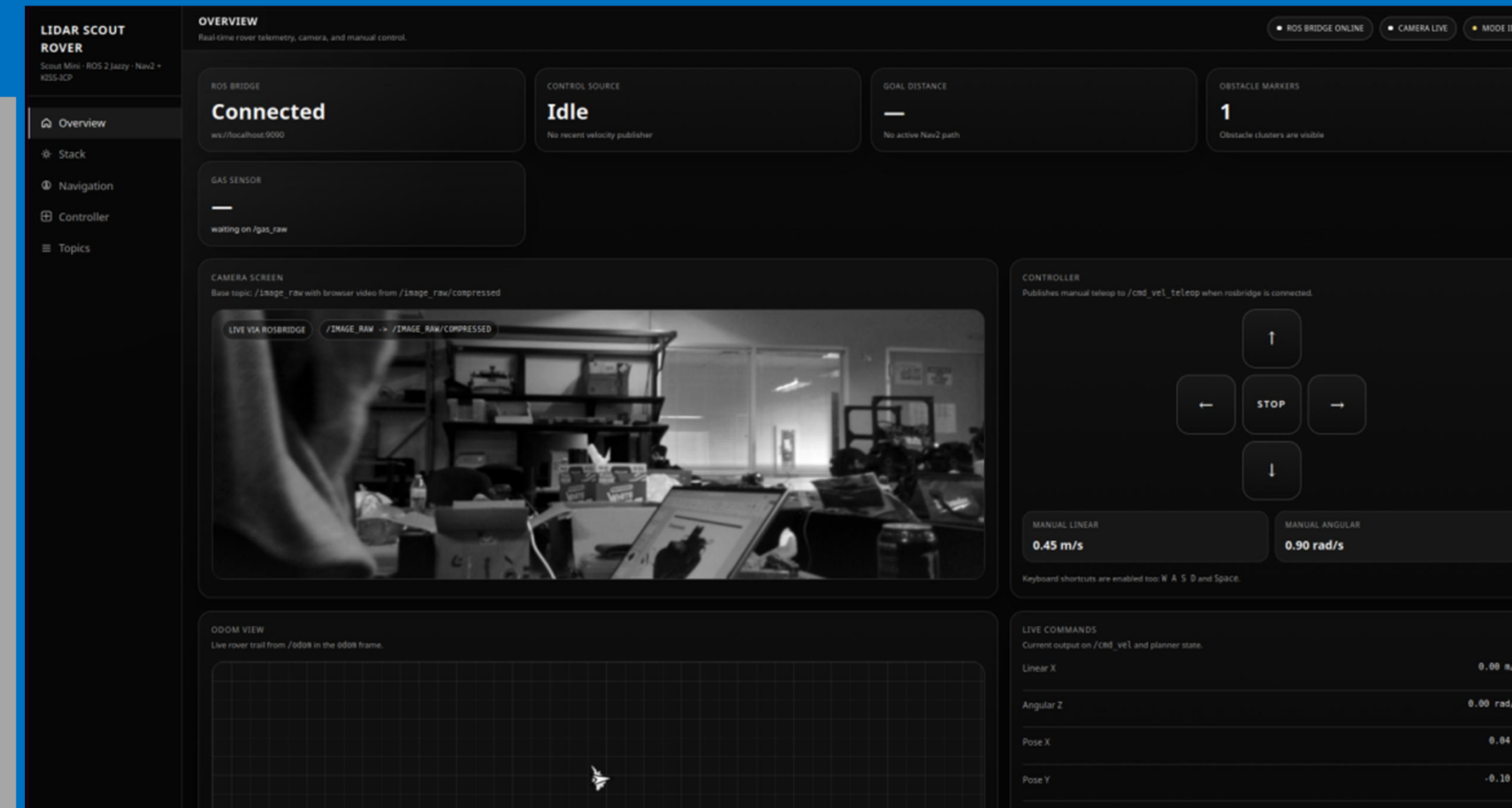
Background:

Hazardous, confined, and GPS-denied environments can be difficult to inspect safely using human personnel alone. Many robotic systems are expensive, privately owned, or limited in adaptability for different field conditions. The LiDAR Scout Rover was developed as a low-cost, ground-based platform for real-time mapping, remote inspection, and environmental monitoring in applications such as disaster response, cave exploration, and building surveying.

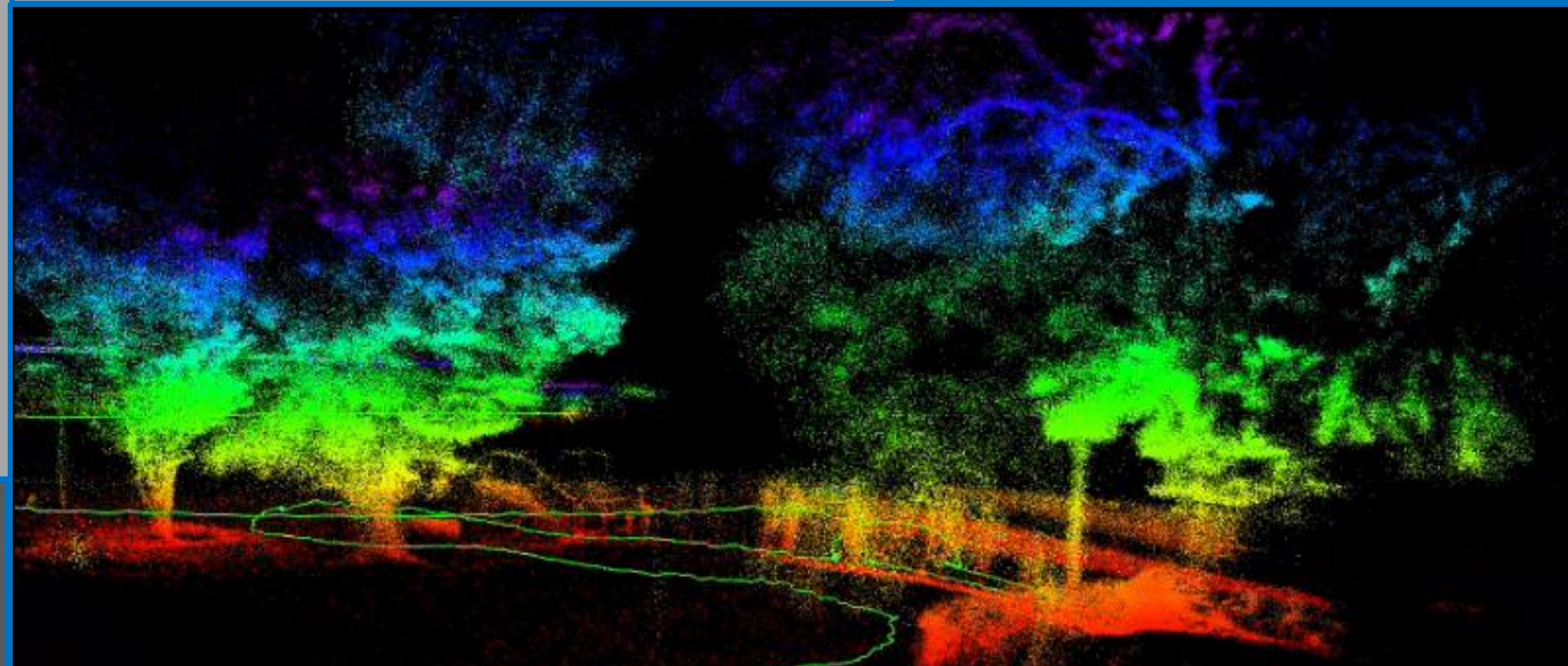
- Uses Unitree LiDAR L1 for real-time mapping
- Uses a webcam and gas sensors for environmental monitoring
- Built on ROS 2 for system communication and control
- Designed for modular use in hazardous and hard-to-access areas
- Provides a more affordable, open-source alternative to commercial systems



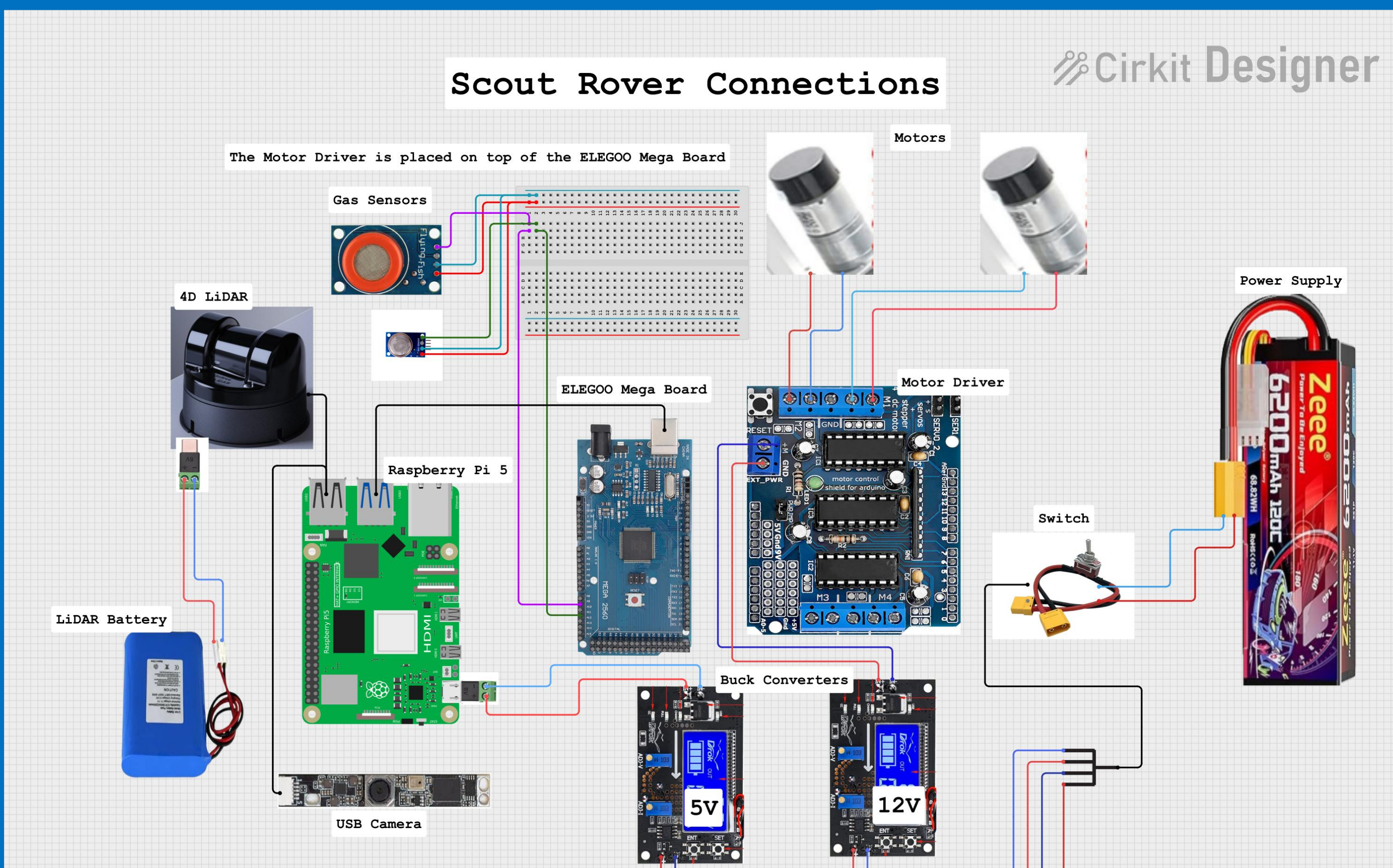
3D rover model in Gazebo simulation



Rover Website UI



RViz2 outdoor mapping with Point-LIO



Electrical and power distribution diagram

Schedule

Semester 1 – Core Rover Development

Research and planning

Proposal, budget, and parts ordering

Chassis assembly and ROS 2 setup

LiDAR, power, and motor control

integration

Semester 2 – Autonomy & Testing

Navigation and path following

Dashboard and connectivity improvements

Webcam/gas sensor integration

Testing and validating Rover

Budget

Component

Cost

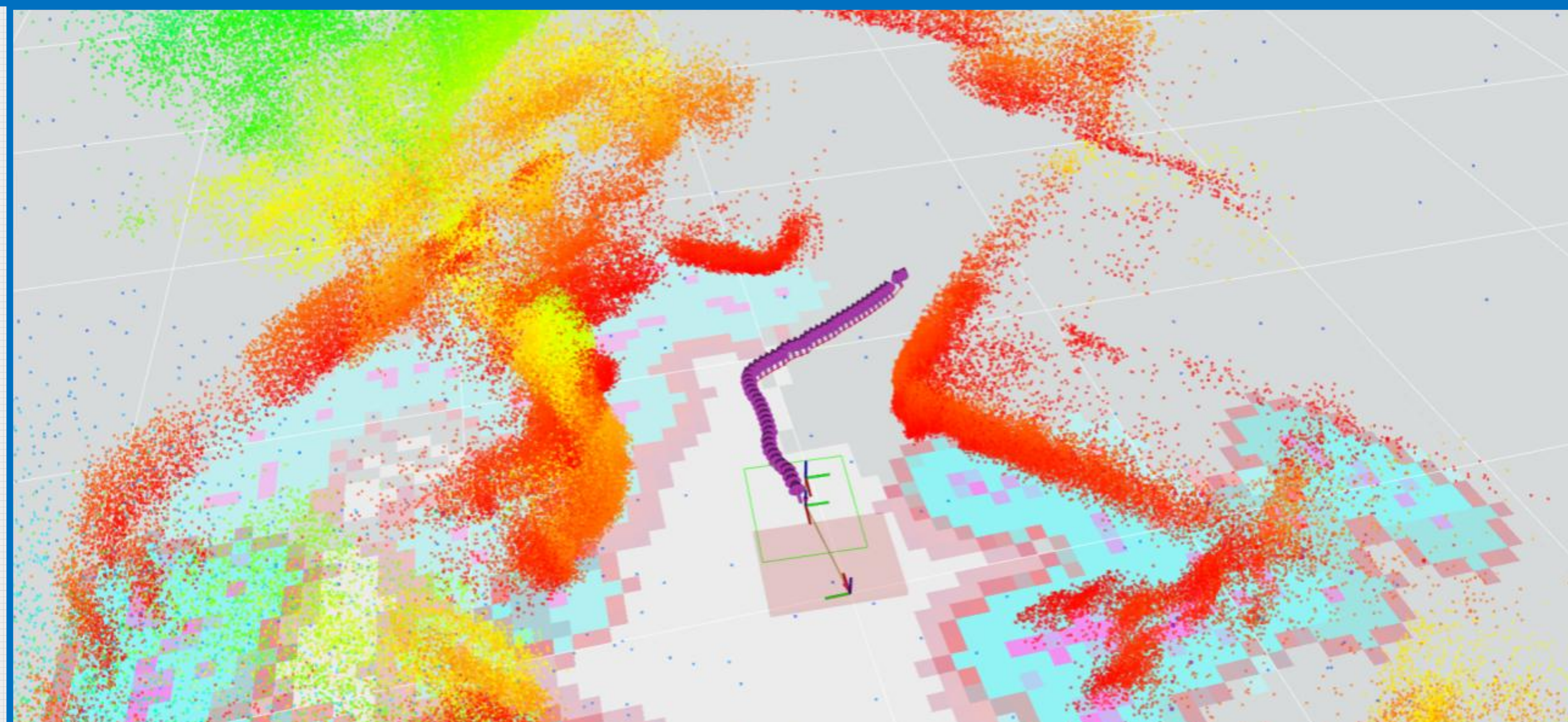
Raspberry Pi 5	\$65
Unitree LiDAR L1	\$250
Arduino Mega	\$45
Camera	\$30
Gas Sensors	\$25
Tracked Chassis	\$100
Motor Driver & Battery	\$112
Wiring & 3D Printed Parts	\$40
Contingency & Incidentals	\$101
Total	\$768

Methodology

- **ROS 2** integrates sensing, autonomy, and rover control in real time.
- **Tracked chassis** with 3D printed parts provides stable mobility across rugged terrain.
- **Point-LIO** performs LiDAR-inertial mapping and visualization on the laptop.
- **Gas sensors** detect hazardous gases in the rover's environment.
- **Web dashboard** provides live telemetry, webcam viewing, and manual control.
- **Nav2** supports autonomous path planning, obstacle avoidance, and goal navigation.



Final LiDAR Scout Rover prototype



Nav2 enables autonomous rover navigation using real-time LiDAR odometry, obstacle detection, and path planning in RViz.